

Assignment 2

In this assignment you are required to write 3 functions:

Q1: A function that simulates the multivariate normal distribution.

Q2: A function that simulates the ARMA(1,1) process.

Q3: A function that simulates compound random variables.

In each one of the functions that are in the file you are required to replace the place-holder “# YOUR CODE HERE” by the body of the code of the function that you write. You must not change the name of the function nor the arguments of the function. Do not add any comments to the code and do not add any code outside the body of the functions. Unless otherwise stated, your code cannot assume availability of or call any package besides the standard packages in the base distribution of R that are uploaded automatically when an R session opens.

You can use the script “Assignment_2_Test.R” in order to test your code. Place this script and your code in the working directory of R and run the expression:

```
> source("Assignment_2_Test.R")
```

Question 1 (Q1):

We introduced in class 3 functions for simulating the multivariate normal distribution: ``mvrnorm`` from the “MASS” package, ``rmvnorm`` from the “mvtnorm” package, and ``rmvn`` function from the “mvnfast” package.

Recall that if $\vec{X}$ is a random vector with independent standard normal components then the linear transformation $\vec{Y} = A\vec{X} + \vec{\mu}$, where A is a deterministic matrix and $\vec{\mu}$ is a deterministic vector, has a multivariate normal distribution with expectation $\vec{\mu}$ and variance covariance matrix $\Sigma = AA^T$. You may use the function ``rnorm`` to generate standard normal random variables and use linear transformation in order to obtain simulated vectors from the required multinormal distribution.

The arguments of the function are:

n = The number of independent row vectors from the multivariate normal distribution.

μ = a vector giving the means of the variables.

Σ = a positive-definite symmetric matrix specifying the covariance matrix of the variables.

The output of the function should be a matrix with n rows and $\text{length}(\mu)$ columns. Each row should correspond to an independent copy of a random variable from the multivariate normal distribution with mean ``mu`` and variance-covariance matrix ``Sigma``.

Remark: You must not use neither of the functions ``mvrnorm``, ``rmvnorm``, nor ``rmvn``.

Question 2 (Q2):

The ARMA(1,1) process is frequently used in the modeling of time series. This process is parametrized by 4 numbers: $-1 < a, b < 1$, and $\sigma, \sigma_0 > 0$. The initiation of the process is with $Y_0 \sim N(0, \sigma_0^2)$, and it is defined via the recursive formula:

$$Y_i = aY_{i-1} + b\epsilon_{i-1} + \epsilon_i, \quad 1 \leq i \leq m.$$

The variables ϵ_i , $0 \leq i \leq n$, are independent $N(0, \sigma^2)$ random variables. In this question you are required to write a function that produces n independent copies of the process $Y_1, Y_2, \dots, Y_m$.

The arguments of the function are:

n = The number of independent copies of the process

m = The length of the process.

a = The AR parameter.

b = The MA parameter.

sd = The standard deviation of the random noise ϵ .

sd.0 = The standard deviation of the initial Y_0 .

The output of the function should be a matrix with n rows and m columns. Each row should correspond to an independent copy of the process $Y_1, Y_2, \dots, Y_m$.

Remark: You must not use the function ``arima.sim``.

Question 3 (Q3):

Compound random variables are frequently for modeling in many statistical applications. For example, in actuary one uses compound random variables in order to model the total claim in a given period.

A compound random variable S is defined with an integer-valued random variable N and a sequence of independent random variables $X_1, X_2, X_3, \dots$ via:

$$S = \sum_{i=1}^N X_i,$$

If $N > 0$. If $N = 0$ then $S = 0$.

Write a function by the name “a2q3” that outputs a sequence of numbers from the compound distribution.

The arguments of the function are:

n = The length of the sequence of random numbers.

rN = a function, provided by the user, that generates integer-valued random numbers.

rX = a function, provided by the user, that generates random numbers X_i that serve as the summands in the definition of the compound random number S .

The function should return a vector of length n of pseudo-random numbers from the compound distribution S .

Remark: You must not use the character string ‘Compound’.